



**International Currency Technologies Inc.**

# **ICT Protocol For RS232**

**Communication Specification**

Change History

| Title     |         | ICT Protocol For Forth                  |                           |        |          |
|-----------|---------|-----------------------------------------|---------------------------|--------|----------|
| Published |         | ICT, Department of Software Development |                           |        |          |
| Item      | Version | Date                                    | History                   | Modify | Comments |
| 1         | 0.1     | 2004.07.30                              | First Edition             | D.C    |          |
| 2         | 0.2     | 2011.01.24                              | Add cmd 0x5A, Escrow Hold | Niki   |          |
| 3         | 0.3     | 2014.01.15                              | Remove 0x26 response      | ACQUA  |          |
|           |         |                                         |                           |        |          |
|           |         |                                         |                           |        |          |
|           |         |                                         |                           |        |          |
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1. General Information

This specification describes the operation of ICT Interface Protocol.  
It is regarding to the data for the interface between Controller and ICT product.

2. Communication Format

|                           |                          |      |
|---------------------------|--------------------------|------|
| Transmission Method       | Full Duplex Transmission |      |
| Transmission Speed        | 9600 Baud Rate           |      |
| Synchronizing Method      | Asynchronous Mode        |      |
| Connection Control Method | Request Method           |      |
| Data Format               | Start bit                | 1    |
|                           | Data bit                 | 8    |
|                           | Parity bit               | Even |
|                           | Stop bit                 | 1    |
|                           | X parameter              | None |

### 3. Commands

#### 3.1 Power Up

| Bill Acceptor                |           |           | Controller |                  |
|------------------------------|-----------|-----------|------------|------------------|
| Category                     | Command   | Direction | Command    | Description      |
| Power Supply ON              | 80H (128) | ➔         |            |                  |
|                              | 8FH (143) |           |            |                  |
| 1. Received 02H in 2 seconds |           | ➔         | 02H (02)   | Within 2 seconds |

\* Bill Acceptor will be Enable status until receive 0x02 from Controller.

#### 3.2 Escrow

| Bill Acceptor                             |                              |           | Controller                                        |                                                                                                                             |
|-------------------------------------------|------------------------------|-----------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Category                                  | Command                      | Direction | Command                                           | Description                                                                                                                 |
| Send bill validated                       | 81H (129)                    | ➔         |                                                   | Bill validated                                                                                                              |
| Send bill value                           | 40H ~ 44H                    | ➔         |                                                   | 40H : First Bill Type<br>41H : Second Bill Type<br>42H : Third Bill Type<br>43H : Fourth Bill Type<br>44H : Fifth Bill Type |
|                                           | Wait for Request (5 Seconds) | ➔         | 02H (02)<br>Or<br>0FH (15)<br>Or<br>18H(24)<br>Or | a. Request to Accept<br>Or<br>b. Request to Reject<br>Or<br>c. *Request to Hold in Escrow<br>d. *No response (5 seconds)    |
| Stacking (for a.)<br>Or<br>Reject (for b) | 10H (16)<br><br>11H (17)     | ➔         |                                                   | Controller do not need to respond                                                                                           |

\*Note 1: 18H will hold the bill in Escrow until receive Accept (0x02h) or Reject (0x0fh).

\*Note 2: If no any response after 5 mins Escrow Timeout, Bill Acceptor will Reject Bill automatically. If it continue 3 Times Escrow Timeout, Bill Acceptor Response Communication Failure.

3.3 Controller Polling Bill Acceptor Status

| Bill Acceptor                                 |          |           | Controller |                                            |
|-----------------------------------------------|----------|-----------|------------|--------------------------------------------|
| Category                                      | Command  | Direction | Command    | Description                                |
| Controller Polling<br>Bill Acceptor<br>Status |          |           |            |                                            |
|                                               |          | ←         | 0CH (12)   | Request Bill Acceptor<br>Status            |
| Restart BA                                    |          |           |            |                                            |
|                                               | 20H (32) |           |            | Motor Failure                              |
|                                               | 21H (33) |           |            | Checksum Error                             |
|                                               | 22H (34) |           |            | Bill Jam                                   |
|                                               | 23H (35) |           |            | Bill Remove                                |
|                                               | 24H (36) |           |            | Stacker Open                               |
|                                               | 25H (37) |           |            | Sensor Problem                             |
|                                               | 27H (39) |           |            | Bill Fish                                  |
|                                               | 28H (40) |           |            | Stacker Problem                            |
|                                               | 29H (41) |           |            | Bill Reject                                |
|                                               | 2AH (42) |           |            | Invalid Command                            |
|                                               | •        |           |            | •                                          |
|                                               | •        |           |            | •                                          |
|                                               | 2EH(46)  |           |            | Reserved                                   |
|                                               | 2FH(47)  |           |            | Response when Error<br>Status is Exclusion |
|                                               | 3EH (62) |           |            | Bill Acceptor Enable<br>Status             |
|                                               | 5EH (94) |           |            | Bill Acceptor Inhibit<br>Status            |

### 3.4 Enable/Disable

| Bill Acceptor |         |           | Controller |                                  |
|---------------|---------|-----------|------------|----------------------------------|
| Category      | Command | Direction | Command    | Description                      |
|               |         | ←         | 3EH (62)   | Controller Enable Bill Acceptor  |
|               |         | ←         | 5EH (62)   | Controller Disable Bill Acceptor |

### 3.5 Reset

| Bill Acceptor                       |                        |           | Controller |             |
|-------------------------------------|------------------------|-----------|------------|-------------|
| Category                            | Command                | Direction | Command    | Description |
| <b>Command Bill Acceptor Reset</b>  |                        | ←         | 30H (48)   |             |
| <b>Power Supply ON</b>              | 80H (128)<br>8FH (143) | →         |            |             |
| <b>1. Received 02H in 2 seconds</b> |                        | ←         | 02H (02)   |             |

\* Bill Acceptor will be Enable status until receive 0x02 from Controller.



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